Instituto Tecnológico y de Estudios Superiores de Occidente

Reconocimiento de validez oficial de estudios de nivel superior según acuerdo secretarial 15018, publicado en el Diario Oficial de la Federación del 29 de noviembre de 1976.

Department of Mathematics and Physics
Master of Data Science

Thesis Title

THESIS to obtain the **DEGREE** of
MASTER OF DATA SCIENCE

A thesis presented by:
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Thesis Advisors:
Dr. Willie E. Coyote
Dr. Homer J. Simpson

Tlaquepaque, Jalisco, January, 2021

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Master of Data Science Approval Form

Thesis Title: **Thesis Title**
Author: **Joey Ramone**
Thesis Approved to complete all degree requirements for the Master of Science Degree in Data Science.

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Tlaquepaque, Jalisco, January, 2021

Thesis Title

Joey Ramone

Abstract

Here

Thesis Title

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Resumen

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1 Introduction

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1.1 Learning from Graph-Structured Data

Data in machine learning often comes in a few familiar forms. Most datasets are tabular: a table of rows and columns in which each row is an independent example described by the same fixed set of features. Other data is organized as images, that is, grids of pixels, or as sequences such as text and audio, whose elements follow one another in a specific order. What these forms share is regularity: each has a fixed, well-defined structure that repeats across every example.

Many real-world systems do not fit any of these shapes. A molecule is a set of atoms connected by chemical bonds, and neither the number of atoms nor the way they are connected is the same from one molecule to the next. A social network is a set of people connected by relationships, where each person may be linked to a different number of others. A citation network is a set of papers that reference one another, each citing a different set of others. These are not tables, images, or sequences; they are networks of entities and the relations between them (Figure 1.1). Data of this kind is naturally described by a *graph*: a set of entities, called *nodes*, together with a set of connections between them, called *edges*. Both nodes and edges can carry their own data, for example in a molecule, each node is an atom labeled with its element (carbon, hydrogen, oxygen, and so on) and each edge is a bond of a particular type (single, double, and so on); in a social network, the nodes are people and the edges are the relationships among them.

Rather than treating a domain as a collection of isolated data points, we describe it in terms of entities and the relationships among them, which means there is an underlying graph whose structure ties those

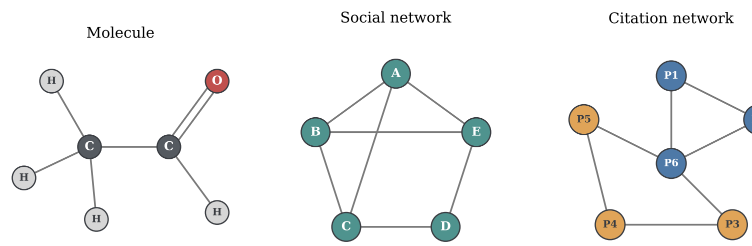

Figure 1.1: Three domains represented as graphs. A molecule, whose nodes are atoms labeled by their element and whose edges are bonds (single or double); a social network, whose nodes are people and whose edges are relationships; and a citation network, whose nodes are papers (colored by topic) and whose edges are citations between them. In each case the entities and their connections, not a table, grid, or sequence, define the data.

entities together. Seen this way, the graph is not merely one possible encoding of a molecule or a social network but a faithful representation: it records exactly which entities exist and how they relate, without forcing the data into a shape it does not have. A wide range of domains exhibit this kind of relational structure, and modeling them explicitly as graphs, rather than ignoring the connections, can lead to more accurate predictions. Data organized in this way is called *graph-structured data*.

Most deep learning models are tailored to the regular structures described above, particularly sequences and images laid out on a grid. Graphs are harder to work with. They have no fixed size, since different graphs contain different numbers of nodes. They have no fixed layout: a pixel in a grid has a clear sense of place, with neighbors above, below, and to either side, whereas a node in a graph has no such spatial arrangement and may be connected to any number of others. And they have no natural ordering, with no first or last node and no canonical point from which to read the rest.

The goal, then, is a model that takes a graph as input and produces predictions about it, such as a label for each node, wheter connections exist between pairs of nodes, or a single label for the graph as a whole. This raises the central question: how can a predictive model – concretely, a neural network – be designed to operate directly on a graph, respecting its variable size and its lack of order instead of discarding them? Neural networks built for this purpose are called *Graph Neural Networks*, hereafter abbreviated as GNNs.

1.2 *Objectives and Scope*

The general objective of this work is to provide a self-contained reference on GNNs for homogeneous graphs under the *message-passing* framework: a single document that develops the theory that motivates and defines these models, presents representative architectures, and evaluates them through reproducible implementations.

Specifically, the work aims to:

- establish the foundations that motivate GNNs: how graphs represent relational data, why standard neural networks cannot learn from it directly, and the inductive biases a model on graphs must respect;
- develop the message-passing framework as the unifying formulation from which GNN architectures are derived;
- present representative architectures for the three standard learning tasks on graphs: the Graph Convolutional Network (GCN) for node classification, the Graph Auto-Encoder (GAE) for link prediction, and the Graph Isomorphism Network (GIN) for graph classification;
- implement the three architectures on standard benchmark datasets (Cora, Amazon Computers, and NCI1) using PyTorch Geometric, with reproducible implementations, and evaluate each against a multilayer perceptron baseline that ignores the graph, examining the predictive benefit of explicitly modeling the relational structure of the data;
- examine the principal limitations of GNNs under this framework, including *oversmoothing* and the ceiling on their expressive power.

Its contribution is the reference document itself: a single, reproducible source that develops the message-passing framework from first principles, presents representative GNN architectures, and serves as a foundational resource for studying the field.

The graphs we consider are treated mainly as *undirected*, meaning their connections carry no direction, and *simple*, meaning that at most one connection joins any pair of nodes and that no node is connected to itself. This choice reflects simplicity rather than a limitation of the framework: message passing has enough design space to accommodate *directed* graphs, *self-loops* (a node connected to itself), and even *multigraphs* (more than one connection between the same pair of nodes).

The graphs considered in this work are also *homogeneous*: they contain a single kind of node and a single kind of connection. This restriction is deliberate. The homogeneous setting is where the message-passing framework is most directly understood, and it is the foundation on which more elaborate models are built, including *heterogeneous* graphs, which carry several kinds of nodes and connections and require machinery beyond the scope of this work. Establishing the homogeneous case is a prerequisite for these extensions, which we leave as future research. The formal definitions of these graph types are given later in the document.

1.3 Structure of the Document

This introduction has set out the motivation, objectives, and scope of the work. The remainder of the document is organized as follows.

The first chapters establish the foundations. Chapter **??** surveys the tasks that graph learning addresses at the node, edge, and graph level, together with the practical applications that motivate the rest of the work. Chapter **??** establishes the formal definitions and notation for graphs used throughout. Chapter **??** reviews the multilayer perceptron and shows why standard neural networks are ill-suited to relational data. Chapter **??** develops the notion of inductive bias and the permutation invariance and equivariance that any graph layer must satisfy.

Chapter **??** builds on these foundations to derive the message-passing framework: the message, aggregation, and update functions, the receptive field that successive layers induce, the expressive power of the resulting models, and the architectural design space it enables.

Chapter **??** presents the three representative architectures within this framework: GCN for node classification, GAE for link prediction, and GIN for graph classification. Chapter **??** turns to practice, providing reproducible implementations of these architectures on the Cora, Amazon Computers, and NCI1 benchmarks in PyTorch Geometric, with the accompanying code available in a companion repository.

Chapter **??** examines the limitations of GNNs under the message-passing framework, and Chapter **??** synthesizes the work and outlines directions for future research.